



FleetFlow

Fleet Management & Logistics Tracking Platform

Project Documentation

Infosys Internship Project

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FastAPI	React 19	PostgreSQL	Docker
Python 3.13	SQLAlchemy	JWT + RBAC	WebSockets

1. Introduction

FleetFlow is a full-stack, enterprise-grade Fleet Management and Logistics Tracking Platform developed as part of the Infosys Internship Program. It provides a centralized, real-time system for managing vehicles, drivers, shipments, fuel, maintenance, and logistics analytics — all secured with JWT-based authentication and role-based access control.

Tech Stack

Layer	Technology
Backend	Python 3.13, FastAPI, SQLAlchemy 2.0
Database	PostgreSQL 16 + Alembic Migrations
Authentication	JWT (python-jose) + bcrypt
Async Tasks	Celery + Redis
Reports	ReportLab (PDF) + OpenPyXL (Excel)
Frontend	React 19, Vite 8, Axios
Real-Time	WebSockets (GPS Tracking)
Deployment	Docker Compose (6 services)

2. Problem Statement

Modern logistics operations suffer from critical inefficiencies due to disconnected systems and manual processes. Fleet managers lack a unified platform to oversee vehicles, drivers, and deliveries in real time, leading to costly breakdowns, missed deliveries, and poor decisions.

Challenge	Business Impact
No Real-Time Vehicle Tracking	Delayed response, poor customer communication
Manual Dispatch & Assignment	Errors, bottlenecks, resource conflicts
Reactive Maintenance	Costly breakdowns, safety risks, downtime
Fragmented Data (spreadsheets)	No single source of truth, data loss
No Analytics or Reporting	Poor decision-making, no KPI visibility
No Alerting System	Critical events (overdue service) go unnoticed

3. Objectives

#	Objective	Description
1	Real-Time Vehicle Tracking	GPS coordinates pushed live via WebSockets to all dashboard clients.
2	Intelligent Dispatch	Create shipments, assign drivers/vehicles with full lifecycle management.
3	Proactive Maintenance	Schedule records, auto-generate alerts (service due / overdue / health critical).

#	Objective	Description
4	Fuel Monitoring	Log every fueling event; analytics on consumption, cost, and vehicle usage.
5	Driver Performance	Track attendance, safety score, trip count, distance, and ratings per driver.
6	Role-Based Access Control	4-tier RBAC: Admin → Fleet Manager → Dispatcher → Driver.
7	Analytics & Reports	Operational KPIs + exportable PDF/Excel reports for all modules.
8	Notifications	Multi-channel (push/email/SMS) alerts with priority levels and read tracking.

4. Workflow

4.1 Authentication Flow

User submits credentials → Backend validates via bcrypt → JWT token issued → Token stored in sessionStorage → All API requests auto-attach token via Axios interceptor → Backend decodes token and enforces role on every request.

Step	Action	Result
1	POST /auth/login	Credentials validated
2	JWT Generated	access_token returned (8h expiry)
3	Token Stored	sessionStorage in browser
4	API Request	Authorization: Bearer <token> attached
5	Role Check	RBAC enforced — 403 if unauthorized

4.2 Shipment Dispatch Lifecycle

Status	Triggered By	Side Effects
pending	Dispatcher creates shipment	Shipment record created
in_transit	Dispatcher assigns driver + vehicle	Driver → unavailable, Vehicle → in_transit, Trip created
delivered	Driver marks delivery complete	Driver → available, Vehicle → available, delivered_at set
cancelled	Dispatcher cancels shipment	Resources freed, status final

State flow: pending → in_transit → delivered | cancelled

4.3 Real-Time GPS Tracking

A background simulation thread starts on server startup and updates each vehicle's lat/lng every 2 seconds. Updates are broadcast via WebSocket to all connected clients (LiveMap page). Drivers can also push manual location updates via **PATCH /gps/vehicles/{id}/location**.

4.4 Maintenance & Alert Workflow

Fleet managers schedule maintenance records per vehicle. Celery Beat periodically scans records and auto-generates alerts for: overdue services, health scores below threshold, and upcoming scheduled services. Alerts appear in the dashboard for acknowledgment and resolution.

4.5 Report Export Workflow

Report Type	Formats	Access
Fleet Summary	PDF, Excel	Admin, Fleet Manager
Driver Performance	PDF, Excel	Admin, Fleet Manager, Dispatcher
Fuel Consumption	PDF, Excel	Admin, Fleet Manager
Maintenance History	PDF, Excel	Admin, Fleet Manager
Shipment / Delivery	PDF, Excel	Admin, Dispatcher

5. Screenshots & UI Overview

FleetFlow features a dark-themed, role-adaptive web UI with 14 screens. Below are representative mockups of the key interface pages.

Screen	Component	Access	Key Features
Login	Login.jsx	Public	JWT auth, role-based redirect
Dashboard	Dashboard.jsx	All roles	7 live KPI stat cards
Live Map	LiveMap.jsx	Admin/FM/Disp	WebSocket GPS tracking, markers
Vehicles	Vehicles.jsx	Admin/FM	CRUD table, GPS coordinates
Drivers	Drivers.jsx	Admin/FM/Disp	Performance metrics, attendance
Driver Assignment	DriverAssignment.jsx	Admin/Disp	Conflict-free assignment
Shipments	Shipments.jsx	Admin/Disp/Driver	Status lifecycle, cancel
Maintenance	Maintenance.jsx	Admin/FM	Health score, scheduling
Maintenance Alerts	MaintenanceAlerts.jsx	Admin/FM	Auto-alerts, acknowledge
Fuel	Fuel.jsx	Admin/FM/Driver	Fuel logs, cost analytics
Notifications	Notifications.jsx	All roles	Priority inbox, multi-channel
Reports	ReportsExport.jsx	Admin/FM/Disp	PDF & Excel downloads
Trips	Trips.jsx	Admin/FM/Disp	Trip tracking & history

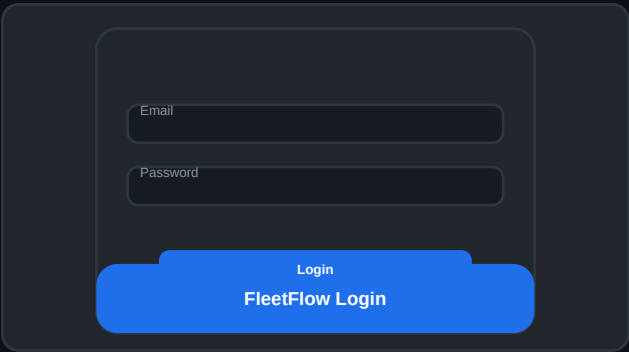


Fig 1: Login Page

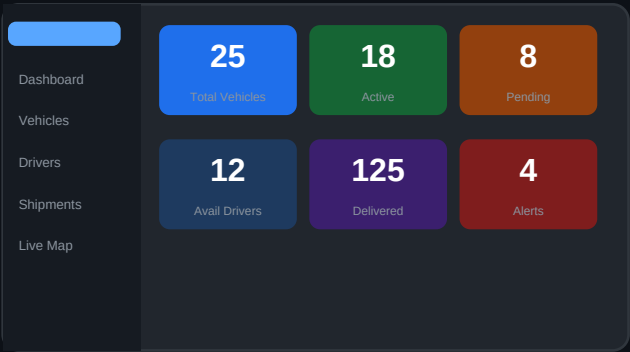


Fig 2: Dashboard

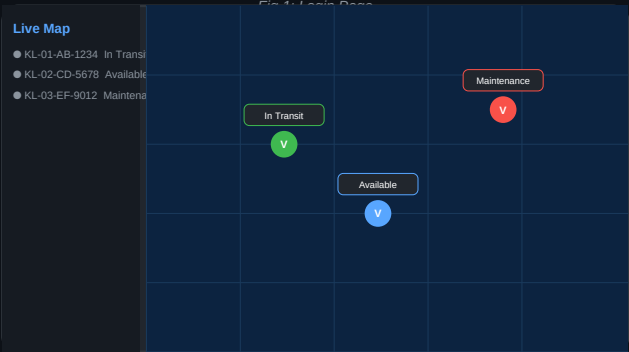


Fig 3: Live Map (GPS)

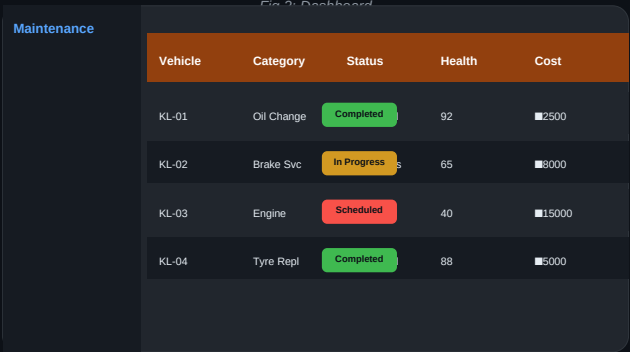


Fig 4: Maintenance

6. Conclusion

FleetFlow successfully delivers a production-ready, end-to-end fleet management solution that consolidates all logistics operations into a single, intelligent platform. The system eliminates manual processes, provides real-time visibility, and empowers each stakeholder role with precisely the tools they need.

Key Achievements

Metric	Value
API Endpoints	70+ REST endpoints + 2 WebSocket channels
Database Tables	10 normalized PostgreSQL tables
Frontend Screens	14 role-adaptive React pages
Docker Services	6 containers (PostgreSQL, Redis, Backend, Celery ×2, Frontend)
Report Formats	PDF (ReportLab) + Excel (OpenPyXL) for 5 report types
Background Tasks	Celery Beat for scheduled maintenance alerts
Security	JWT RBAC with 4 roles, bcrypt passwords, session-scoped tokens

Challenges & Solutions

Challenge	Solution Applied
bcrypt/passlib compatibility	Used bcrypt library directly, bypassing passlib
Real-time GPS without hardware	Background simulation thread broadcasting via WebSocket
N+1 query performance	SQLAlchemy aggregate functions (func.sum, func.count)
Role-scoped data access	RBAC dependency injection — enforced at middleware layer

Future Enhancements

- Google Maps integration for real geocoding and route visualization
- Email/SMS notification delivery via SendGrid / Twilio
- React Native mobile app for on-the-go driver tracking
- AI-powered route optimization and predictive maintenance
- Customer shipment tracking portal with public links
- Multi-tenant support for multiple fleet organizations

"FleetFlow transforms fragmented fleet data into real-time operational intelligence, empowering logistics teams to move faster, smarter, and safer."

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